Worksheet 35 · Sine & Cosine Rules IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use the sine rule to find missing sides and angles • Use the cosine rule to find missing sides and angles • Find the area of a triangle using ½ab sin C		
Grade: / 61	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

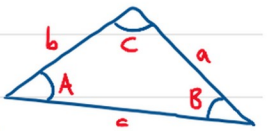
Sine for Sides → $\frac{a}{\sin A} = \frac{b}{\sin B}$

Sine for Angles → $\frac{\sin A}{a} = \frac{\sin B}{b}$

Cosine for Sides → $a^2 = b^2 + c^2 - 2bc\cos A$

Cosine for Angles → $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

Area of Triangle → $A = \frac{1}{2}ab\sin C$



Questions *Sine & Cosine Rules*

Question 1 (3 marks)

The diagram shows triangle PQR.

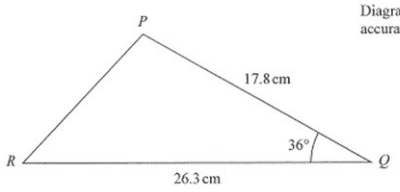


Diagram **NOT** accurately drawn

Calculate the length of PR.
Give your answer correct to 3 significant figures.

..... cm

Question 2 (3 marks)

A triangle has sides of length 8 cm, 10 cm and 14 cm.

Work out the size of the largest angle of the triangle.
Give your answer correct to 1 decimal place.

Question 3 (5 marks)

Here is triangle ABC.

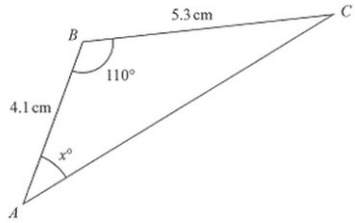


Diagram **NOT** accurately drawn

Calculate the value of x.
Give your answer correct to 3 significant figures.

Question 4 (5 marks)

ABCD is a quadrilateral.

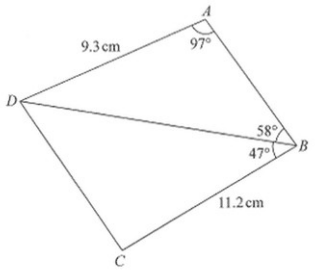


Diagram **NOT** accurately drawn

Find the area of quadrilateral *ABCD*.
Give your answer correct to 3 significant figures.

..... cm²

Question 5 (5 marks)

Here is a triangle *XYZ*.

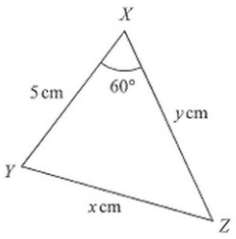


Diagram **NOT** accurately drawn

The perimeter of the triangle is *k* cm.

Given that $x = y - 1$
find the value of *k*.
Show your working clearly.

k = ...

Question 6 (3 marks)

The diagram shows a kite *ABCD*

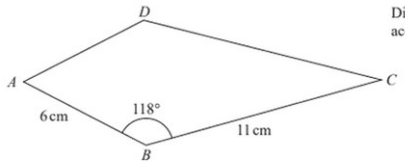


Diagram **NOT** accurately drawn

AB = 6 cm *BC* = 11 cm Angle *ABC* = 118°

Calculate the area of the kite.
Give your answer correct to 3 significant figures.

..... cm²

Question 7 (6 marks)

ABCD is a quadrilateral.

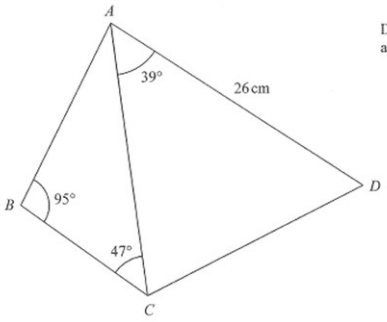


Diagram **NOT** accurately drawn

The area of triangle *ACD* is 250 cm²

Calculate the area of the quadrilateral *ABCD*.
Show your working clearly.
Give your answer correct to 3 significant figures.

Question 8 (5 marks)

Here is triangle *ABC*

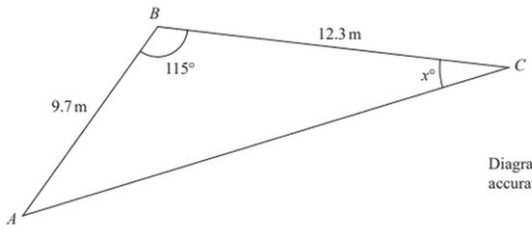


Diagram **NOT** accurately drawn

Work out the value of *x*
Give your answer correct to 3 significant figures.

Question 9 (4 marks)

Here is a parallelogram *PQRS*, in which angle *SPQ* is acute.

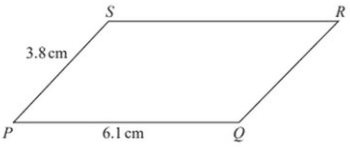


Diagram **NOT** accurately drawn

PQ = 6.1 cm *PS* = 3.8 cm

The area of the parallelogram is 18 cm²

Work out the length of *QS*
Give your answer correct to 3 significant figures.

Question 10 (6 marks)

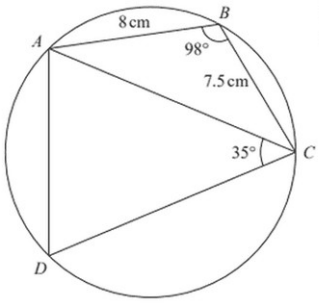


Diagram **NOT** accurately drawn

ABCD is a quadrilateral where *A*, *B*, *C* and *D* are points on a circle.

AB = 8 cm
BC = 7.5 cm
Angle *ABC* = 98°
Angle *ACD* = 35°

Work out the perimeter of quadrilateral *ABCD*.
Give your answer correct to one decimal place.

Question 11 (5 marks)

Here is a quadrilateral *ABCD*.

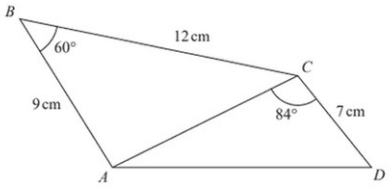


Diagram **NOT** accurately drawn

Calculate the area of quadrilateral *ABCD*.
Give your answer correct to 3 significant figures.
Show your working clearly.

Question 12 (5 marks)

The diagram shows triangle *PQR*

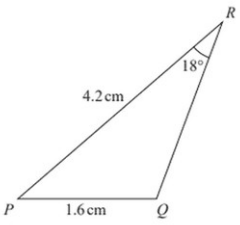


Diagram **NOT** accurately drawn

PQ = 1.6 cm *PR* = 4.2 cm Angle *PRQ* = 18°

Given that angle *PQR* is obtuse,

work out the area of triangle *PQR*
Give your answer correct to 3 significant figures.

Question 13 (6 marks)

The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

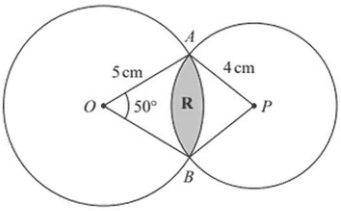


Diagram **NOT** accurately drawn

One of the circles has centre *O* and radius 5 cm.
The other circle has centre *P* and radius 4 cm.
Angle *AOB* = 50°

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.